

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM
January 2012
ALGEBRA (Ph.D. Version)

Instructions to the student

- a. Answer all six questions; each will be assigned a grade from 0 to 10
 - b. Use a different booklet for each question. Write the problem number and your **code number** (not your name) on the outside of the booklet.
 - c. Keep scratch work on separate pages in the same booklet.
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1. Let G be a finite group and let p be the smallest prime dividing $|G|$. Suppose that the Sylow p -subgroup H of G is normal and cyclic. Show that H lies in the center of G (the center is the set of elements that commute with all elements of G . *Hint:* What is the order of the automorphism group of $\mathbf{Z}/p^n\mathbf{Z}$?)
2. Let $S \subset M_n(\mathbf{C})$ be a set of $n \times n$ complex matrices such that $AB = BA$ for all $A, B \in S$. Assume that every matrix in S is nilpotent (that is, $A^n = 0$). Show that there exists a matrix M such that MAM^{-1} is upper triangular for all $A \in S$ (that is, all entries on or below the diagonal are 0; *Note:* M is independent of A).
3. Find all prime ideals of the ring $\mathbf{Z}[T]/(T^2)$
4. Let R be a commutative ring with 1 and let A, B, C, D, E be R -modules. Suppose we have a commutative diagram

$$\begin{array}{ccccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & E & \longrightarrow & 0 \\ \downarrow \alpha & & \downarrow \beta & & \downarrow = & & \\ C & \xrightarrow{r} & D & \xrightarrow{s} & E & \longrightarrow & 0 \end{array}$$

where the maps are R -module homomorphisms and the rows are exact

- (a) Show that r induces a well-defined map $\bar{r} : C/\alpha(A) \rightarrow D/\beta(B)$.
- (b) Show that $\bar{r}$ is surjective.
- (c) Suppose that r is injective. Show that $\bar{r}$ is injective.

(*Note:* You may not quote the Snake Lemma, the 4-Lemma, or similar results, since the present result is almost a special case of one of these.)

5. Let ζ be a primitive 7th root of unity in $\mathbf{C}$. Let $\alpha = \zeta + \zeta^{-1}$.
 - (a) Show that $\alpha \notin \mathbf{Q}$.
 - (b) Show that $[\mathbf{Q}(\alpha) : \mathbf{Q}] = 3$.
 - (c) Show that if $\beta \in \mathbf{Q}(\alpha)$ satisfies $\beta^3 \in \mathbf{Q}$ then $\beta \in \mathbf{Q}$.
6. Let G be a finite group, let $\rho : G \rightarrow GL(V)$ be a finite-dimensional representation of G on some complex vector space V , and let $\chi = \text{Trace}(\rho)$ be the character of ρ .

- (a) Let $g \in G$. Show that the eigenvalues of $\rho(g)$ are roots of unity and that $\chi(g^{-1}) = \overline{\chi(g)}$ (= the complex conjugate of $\chi(g)$)
- (b) Suppose that G has odd order, $\chi(g) \in \mathbf{Z}$ for all $g \in G$, and V has odd dimension. Show that

$$\sum_{g \in G} \chi(g)$$

is odd (and therefore nonzero).

- (c) Suppose that G has odd order, $\chi(g) \in \mathbf{Z}$ for all $g \in G$, and V has odd dimension. Show that there is a nonzero vector $v \in V$ such that $\rho(g)(v) = v$ for all $g \in G$